Towards Bayesian joint modelling of ranks and scores to reduce uncertainty in the assessment of grant proposal quality

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Organization

Timeline: From the 2nd of march 2026 to the 5th of October 2026
Schedule: Full time, with one month break in July.
Intermediate presentation: 04.06.2026 from 15:00 to 16:00
Master exam: 26.10.2026 from 13:00 to 14:00

Background and related works

When resources are limited, decisions have to be made on how to allocate them. In research, funding agencies make these decisions by awarding or rejecting grant proposals. Agencies contact panels of experts which individually review, rank or score proposals, the best of which are funded and the rest rejected (Harman, 1998). There is evidence for bias in this process. For instance, funding for innovative, risky or interdisciplinary research is less likely to be awarded than for conservative and simple proposals (Guthrie et al., 2017). Moreover, assessments of individual experts on a panel often do not agree, even after panel meetings, making it hard to objectively assess which proposals are worth funding (Fogelholm et al., 2012).

Despite these shortcomings, there is no alternative to this expert evaluation system. To mitigate its downsides, statistical methods can be used for fairer decision making. Depending on the assessment system, the experts either make qualitative or quantitative statements about the proposals. This project will focus on statistical methods to analyse quantitative statements. They can be separated into three groups depending on the expert evaluation system: (1) The analysis of scores, (2) the analysis of ranks and (3) a combined analysis of score and rank. When proposals are given scores, they are supposed to be evaluate independently of each other. This can be seen as an evaluation on an absolute scale. In opposition, when proposals are given ranks, they are evaluated against each other. This can be seen as an evaluation on a relative scale.

In the analysis of scores, each expert gives a grade to a selection of proposals. These results are then summarised and used to calculate a final ranking. Bayesian methods have been implemented in this context, allowing the funding agencies not only to obtain a summarised rank but also its uncertainty. This allows for better interpretation of the final ranking (Heyard et al., 2022). The main disadvantage of using scores for ranking is that scores can be unreliable and affected by unpredictable external factors. Depending on the expert's previous experiences, preferences, tiredness and mood, the harshness of their evaluation can substantially vary between proposals.

Giving ranks to proposals instead of scores partially solves this issue. Rankings require direct comparison of proposals, this makes them supposedly more reliable than scores (i.e. they are less affected by external factors). However, producing a ranking is often more time consuming for the experts because the comparisons of the proposals can require multiple readings in order to form an opinion on the final ranking. For this reason, it is seldom used on its own

for grant proposal evaluation.

To combine the advantages of both ranking and scoring - the absolute information given by the scores and the relative information provided by the ranks - new combined analysis methods have been developed. These methods use scores assigned by experts and partial ranking of a small sample of proposals to provide estimations of grant quality. This is the case of the Mallows-Binomial model developed by Pearce and Erosheva (2022). This model enables simultaneous assessment of both scores and ranks, but only provides uncertainty based on frequentist approaches. I plan to extend the Bayesian model created in Heyard et al. (2022) and combine it with ranking data as done in the Mallows-Binomial model. This creates a Bayesian model combining both scores and ranks, while providing uncertainty in the final proposed rankings.

Objectives

This project aims at improving statistical methods for proposal ranking by combining evidence on the quality of the proposals generated by scores and ranks. We hypothesise that this would provide more precise information for decision-makers; enabling them to allocate their resources more fairly at no increased cost.

Secondarily, as a student, I will be able to explore, and further develop my understanding of Bayesian methodology by exploring both Laplace and Markov-chain based methods to posterior likelihood estimation.

Methodology

The first part of this thesis will focus on describing, reproducing and comparing existing ranking methods using both scores and ranks. This will include a minimum of 4 methods: (1) summarising scoring data using means and standard deviations (Harman, 1998), (2) analysing ranking data using Mallows model (Mallows, 1957), (3) analysing scores using Bayesian hierarchical mehods (Heyard et al., 2022), and (4) combining scores and ranks using the Malows-Binomial model (Pearce & Erosheva, 2022). More relevant methods identified during literature review may be added to this first part.

Then the focus will shift to developing a Bayesian joint model for combining scores and ranks (eponym to the thesis title). The model will be implemented using Integrated Nested Laplace Approximations (INLA). INLA enables approximative Bayesian inference while remaining computationally efficient (Rue et al., 2016). Moreover, an INLA package implementing Bayesian joint likelihood modelling already exists (Rustand et al., 2024).

To achieve this joint modelling, we will translate the implementation from Heyard et al. (2022) from the JAGS package to the INLA package. From this new implementation, we will use the likelihood of the parameter θ_i, the quality of the grant proposal based on the scoring data (calculated from y_{ij} the scores given to proposal i by expert j) as defined in the paper by Heyard et al. (2022) and combine it with the likelihood of a new parameter γ_i quality of the grant proposal based on ranking data (calculated from ϕ_{ij}, the rank given to proposal i by expert j). The inference of γ_i will be defined during the thesis itself. The calculation of this likelihood will involve Bayesian imputation methods as ranking data is often only available for

a subset of proposals, e.g., the top six or top ten rankings of each expert. Using both ϕ_{ij} and y_{ij} and combining them in the final joint likelihood model, we hope to model the quality of each proposal more precisely than the current state of the art methods.

Data and software

In order to reproduce, compare, and test the various models described in this project, we will use data from Gallo et al. (2023). All analysis will be performed in the programming language `R`. The reports and presentations will be created using the `KnitR` framework. Reproducible code will be made available online on github with the thesis at submission.

No kind of Artificial Intelligence based tools will be used to generate, proofread or design any content in this thesis. This includes all code, text, figures and presentations.

Intermediate steps

- Review literature on statistical methods for analysing ranking and score data.
- Describe, implement, reproduce and compare the methods identified in the literature review. To this end, the data from Gallo et al. (2023) will be analysed. This will include at minimum 4 methods:
    1. Analysing scoring data using means and standard deviations (Harman, 1998).
    2. Analysing scoring data using hierarchical Bayesian models (Heyard et al., 2022).
    3. Analysing ranking data using the Mallows model (Mallows, 1957).
    4. Jointly analysing score and ranking data using the Mallows-Binomial model (Pearce & Erosheva, 2022).
- Describe the methods used in the JAGS and INLA `R` packages.
- Translate the code and functions from the `ERforResearch` package from a JAGS implementation to a INLA implementation (Heyard et al., 2022).
- Describe and implement a Bayesian imputation model to fill incomplete ranking data.
- Define and implement a joint modelling method for ranks and scores using the INLA package (if time allows).
- Compare the results of the models identified in the literature review and the joint model (if time allows).

Expected deliveries (old) - March start

The deliveries are expected at the first meeting of each month.

1. **April:** Draft of literature review, description of method (1) and (2) and implementation thereof on the Gallo et al. (2023) data.
2. **May:** Literature review, description, implementation and comparison of methods (1) to (4).
3. **June:** Description of Markov Chain Monte Carlo (MCMC) and integrated nested Laplace (INLA) methods for posterior likelihood approximation.
4. **July:** `ERforResearch` package translated to the INLA framework.
5. **September:** Description and implementation of Bayesian imputation methods for ranking data.
6. **October:** Full thesis submission.

Expected deliveries (new) - Mid-april start

1. **May:** Draft of literature review, description of method (1) and (2).
2. **June:** Literature review, description, implementation and comparison of methods (1) to (4) on the Gallo et al. (2023) data.
3. **July:** Description of Markov Chain Monte Carlo (MCMC) and integrated nested Laplace (INLA) methods for posterior likelihood approximation.
4. **September:** `ERforResearch` package translated to the INLA framework.
5. **October:** Description and implementation of Bayesian imputation methods for ranking data.
6. **November:** Full thesis submission.

References

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